

Scalability & Future Plan

Date	15 March 2024
Team ID	SWTID-2026-2168
Project Name	EduGenie Google gemini powered learning assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited concurrent user handling	Slow performance during peak hours	High
2	Dependency on external API limits	Potential service downtime/delays	High
3	Lack of online access	Users cannot study without internet	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Basic handling	Implement Load Balancers to distribute traffic
2	Data Storage	Local/Single Database	Migrate to Cloud-based scalable Database (e.g., Firebase/AWS RDS)
3	Performance	Standard Response Time	Implement Caching (Redis) to reduce latency
4	Security	Basic Authentication	Integrate OAuth 2.0 and advanced API encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Multi-language Support	2026	Wider accessibility for global users
Phase 2	Personalized AI Tutor	2026	
Phase 3	Offline Mode & Mobile App	2027	